



Chap.2 Assemblers

수업 목표

- 단순 SIC Assembler의 기본 개념 및 기능에 대해 이해한다.
- 예제를 통해 기본 개념과 기능 학습을 확인한다.

Object Code of SIC Program (Fig 2.2)

5	1000	COPY	START	1000	
10	1000	FIRST	STL	RETADR	141033
15	1003	CLOOP	JSUB	RDREC	482039
20	1006		LDA	LENGTH	001036
25	1009		COMP	ZERO	281030
30	100C		JEQ	ENDFIL	301015
35	100F		JSUB	WRREC	482061
40	1012		J	CLOOP	3C1003
45	1015	ENDFIL	LDA	EOF	00102A
50	1018		STA	BUFFER	0C1039
55	101B		LDA	THREE	00102D
60	101E		STA	LENGTH	0C1036
65	1021		JSUB	WRREC	482061
70	1024		LDL	RETADR	081033
75	1027		RSUB		4C0000
80	102A	EOF	BYTE	C'EOF'	454F46
85	102D	THREE	WORD	3	000003
90	1030	ZERO	WORD	0	000000
95	1033	RETADR	RESW	1	
100	1036	LENGTH	RESW	1	
105	1039	BUFFER	RESB	4096	

Object Code of SIC Program (Fig 2.2)

※ Addresses 1033-2038 is simply reserved by the loader for use by the program during execution

110		.			
115		.	SUBROUTINE TO READ RECORD INTO BUFFER		
120		.			
125	2039	RDREC	LDX	ZERO	041030
130	203C		LDA	ZERO	001030
135	203F	RLOOP	TD	INPUT	E0205D
140	2042		JEQ	RLOOP	30203F
145	2045		RD	INPUT	D8205D
150	2048		COMP	ZERO	281030
155	204B		JEQ	EXIT	302057
160	204E		STCH	BUFFER, X	549039
165	2051		TIX	MAXLEN	2C205E
170	2054		JLT	RLOOP	38203F
175	2057	EXIT	STX	LENGTH	101036
180	205A		RSUB		4C0000
185	205D	INPUT	BYTE	X' F1 '	F1
190	205E	MAXLEN	WORD	4096	001000

Object Code of SIC Program (Fig 2.2)

195		.			
200		.	SUBROUTINE TO WRITE RECORD FROM BUFFER		
205		.			
210	2061	WRREC	LDX	ZERO	041030
215	2064	WLOOP	TD	OUTPUT	E02079
220	2067		JEQ	WLOOP	302064
225	206A		LDCH	BUFFER, X	509039
230	206D		WD	OUTPUT	DC2079
235	2070		TIX	LENGTH	2C1036
240	2073		JLT	WLOOP	382064
245	2076		RSUB		4C0000
250	2079	OUTPUT	BYTE	X'05'	05
255			END	FIRST	

A Simple SIC Assembler

□ Example of Instruction Assemble

STCH BUFFER, X 549039



$(1010100)_2$ 1 $(001)_2$ $(0000)_2$ $(0011)_2$ $(1001)_2$

$(54)_{16}$

$(9)_{16}$

$(039)_{16}$

명령어는 부록의 모든 SIC/XE 명령어에 대한
OP 코드를 참조하면 된다.

A Simple SIC Assembler

- (1), (3), (4), and (5) can be easily accomplished

- (2)

10	1000	FIRST	STL	RETADR	141033
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→ *Forward reference*

- reference to a label that is defined later in the program
- not know the address that will be assigned to RETADR

→ *2 pass assembler*

- it scans source program in twice

- 2 pass assembler

- *First pass*

- Scan the source program for label definitions and assign addresses

- *Second pass*

- Perform most of actual translation previously described

A Simple SIC Assembler

- Forward Reference

- Reference to a label that is defined later in the program

<u>Loc</u>	<u>Label</u>	<u>OP Code</u>	<u>Operand</u>
1000	FIRST	STL	RETADR
1003	CLOOP	JSUB	RDREC
...	...	...	...
1012		J	CLOOP
...	...	...	...
1033	RETADR	RESW	1



문장 번역이 위에서 아래로 한 줄 씩 번역되는
과정에서 아직 정의되지 않은 주소를 가지는
라벨은 접근할 수 없다.

A Simple SIC Assembler

- **Assembler directives (or pseudo-instructions)**
 - Not translated into machine instructions
 - Provide instructions to the assembler itself
 - E.g.
 - **BYTE, WORD**
 - direct the assembler to generate constants as part of the object program
 - **RESB, RESW**
 - instruct the assembler to reserve memory locations without generating data values
 - **START, END**
 - specify the starting/end address for the program

Object Program Format

- Contains three of records
 - Header
 - contain the program name, starting address, and length
 - Text
 - contain translated instructions and data of the program
 - End
 - mark the end of the object program and specify the address in the program where execution is to begin

Object Program Format

- Header record

- Col. 1 H
- Col. 2~7 Program name
- Col. 8~13 Starting address of object program (Hex)
- Col. 14~19 Length of object program in bytes (Hex)

- Text record

- Col. 1 T
- Col. 2~7 Starting address for object code in this record (Hex)
- Col. 8~9 Length of object code in this record in bytes (Hex)
- Col. 10~69 Object code in Hex (2 column per byte)

- End record

- Col. 1 E
- Col. 2~7 Address of first executable instruction in object (Hex)

Object Program (Fig 2.3)

HCOPY 00100000107A
^ ^ ^

T0010001E1410334820390010362810303010154820613C100300102A0C103900102D
^ ^ ^ ^ ^ ^ ^ ^ ^ ^ ^

T00101E150C10364820610810334C0000454F46000003000000
^ ^ ^ ^ ^ ^ ^

T0020391E041030001030E0205D30203FD8205D2810303020575490392C205E38203F
^ ^ ^ ^ ^ ^ ^ ^ ^ ^

T0020571C1010364C0000F1001000041030E02079302064509039DC20792C1036
^ ^ ^ ^ ^ ^ ^ ^ ^

T002073073820644C000005
^ ^ ^

E001000
^

Two passes assembler

■ **Pass 1** (define symbols)

1. Assign addresses to all statements (**generate LOC**)
2. Save the values (**addresses**) assigned to **all labels** for use in Pass 2
3. Perform some processing of **assembler directives** (address assignment, such as the length of data areas)

■ **Pass 2** (assemble instructions and generate object program)

1. Assemble instructions (translating operation codes & looking up addresses)
2. Generate data values defined by BYTE, WORD, etc
3. Perform processing of assembler directives not done during Pass 1
4. Write the object program and the assembly listing

이 동영상 교과목 자료는
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